CareConnect Desktop Accessibility Conformance Report

WCAG 2.1 Edition (Based on VPAT® Version 2.5)

Version 1.6 (Final), November 3, 2026

Name of Product/Version: CareConnect Desktop 0.1.0 (Electron 39 shell around the CareConnect React 19 web app), final build from dev at 176498d, delivered for macOS as CareConnect-macOS-arm64.dmg.

Report Date: November 3, 2026

Product Description: A desktop companion for adults with ADHD and their caregivers: daily schedule, medication tracking with dose actions, appointments, caregiver notes and an activity log, with a native menu bar and keyboard shortcuts.

Contact Information: SWEN 661 Team 5: Shayne McPherson, Quinton Coleman, Antonio Wilson. Repository: github.com/shaynemcp/careconnect-adhd

Notes: Platforms evaluated: Windows 11 (keyboard-only checklist, NVDA 2026.1.1) and macOS (VoiceOver: macOS 26.6.2 on 10/13/2026, macOS 26.3.1 on 11/3/2026). Version 1.6 is the final report. It records the final build and PRs through #102, and corrects how VoiceOver reads the error summary. Ratings do not change: 40 Supports, 2 Partially Supports, 8 Not Applicable. AI was used to help format this report and to script the key presses in the VoiceOver passes. The VoiceOver results come from VoiceOver itself (its log and caption panel); the other results come from the team's own testing.

Evaluation Methods Used:

- Keyboard-only testing of every page, the menu bar, dialogs and shortcuts: the A8 pass (10/4 to 10/6/2026); the A9 keyboard-only checklist on dev 8a7a1a5 (Antonio Wilson, Windows 11, 10/13/2026: 38 pass, 6 fail; PR #75); and a re-test of #70, #73 and #74 after PR #76 (10/13/2026, all pass).
- Screen readers: NVDA 2026.1.1 on Windows 11 with a speech log (10/6/2026), and VoiceOver on macOS on dev 0ff70f6 (10/13/2026: 26 pass, 5 partial, 1 fail, 14 not run; findings SR-1 to SR-10), with a 3:51 video. SR-1, SR-2, SR-4 and SR-10 were re-checked with VoiceOver on PR #79, and SR-3, SR-5 to SR-9, #71 and #72 on 11/3/2026 (screen reader report, section 12). In the 11/3 re-check a script sent the keys and read VoiceOver's log; nobody listened to the audio.
- axe-core 4.14.0 inside the Electron window, in normal and High Contrast mode: 0 violations in 36 scans of 18 screens on dev 0ff70f6 (10/13/2026, docs/accessibility/axe/). The final web build that Electron loads was re-scanned on 11/3/2026; see the Web report.
- Automated tests on the final code (11/3/2026): web 210 tests, 90.86% line coverage; desktop 64 tests, 99.12% line coverage. They include jest-axe on 8 screens, heading order on every main screen, accessible names, High Contrast computed colors, window titles and dialog focus.
- Contrast: npm run check:contrast on the design tokens, run in CI. 200% zoom. Reflow at 320 CSS px and WCAG text spacing on 12 pages (10/9/2026).
- Code review. Not used: Windows contrast themes.

Applicable Standards/Guidelines

This report covers the degree of conformance for the following accessibility standard/guidelines:

Standard/Guideline	Included In Report
Web Content Accessibility Guidelines 2.1	Level A: Yes. Level AA: Yes. Level AAA: No

Terms

The terms used in the Conformance Level information are defined as follows:

- Supports: The functionality of the product has at least one method that meets the criterion without known defects or meets with equivalent facilitation.
- Partially Supports: Some functionality of the product does not meet the criterion.
- Does Not Support: The majority of product functionality does not meet the criterion.
- Not Applicable: The criterion is not relevant to the product.
- Not Evaluated: The product has not been evaluated against the criterion. This can only be used in WCAG Level AAA criteria.

Summary (50 criteria): Supports 40; Partially Supports 2; Does Not Support 0; Not Applicable 8; Not Evaluated 0.

WCAG 2.1 Report

Ratings and remarks apply to the Electron desktop app on Windows 11 and macOS. Issue and pull request numbers refer to github.com/shaynemcp/careconnect-adhd. SR-1 to SR-10 are the findings in docs/accessibility/desktop-screen-reader-report.md.

Table 1: Success Criteria, Level A

Criteria	Conformance Level	Remarks and Explanations
1.1.1 Non-text Content	**Supports**	Decorative icons are hidden from assistive technology (aria-hidden), and icon-only buttons have names: VoiceOver read "Sign out of CareConnect, button" and "Switch role to care recipient, button". Memories photos use the card title as alt text. No icon was read on its own in the VoiceOver pass. axe-core 4.14.0 in the Electron window: 0 violations on 18 screens in normal and High Contrast mode, including image-alt and button-name (dev 0ff70f6, docs/accessibility/axe/, #63).
1.2.1 Audio-only and Video-only (Prerecorded)	**Not Applicable**	The app contains no prerecorded audio or video. The course demo and screen reader videos are submission artifacts, not part of the product.
1.2.2 Captions (Prerecorded)	**Not Applicable**	The app contains no prerecorded audio or video.
1.2.3 Audio Description or Media Alternative (Prerecorded)	**Not Applicable**	The app contains no prerecorded video.

Criteria	Conformance Level	Remarks and Explanations
1.3.1 Info and Relationships	**Partially Supports**	Headings, landmarks, lists, form labels and dialogs are exposed to assistive technology. The VoiceOver rotor lists one h1 per screen with no skipped levels (Home: h1 "Here's your day, Margaret", h2 "Later today", an h3 for each item, h2 "Quick actions") and the banner, main, navigation and contentinfo landmarks. Heading order is tested on every main screen (a11y.test.tsx; #50, PR #59, PR #68). Exception: badge text has no separator, so NVDA reads "Coming upAppointment" (#52, open; VoiceOver reads the two as separate items). The caregiver Dashboard's Alerts heading, which VoiceOver read in fragments (SR-1), is fixed in PR #79: VoiceOver now reads "heading level 2, Alerts, 5 total". Every schedule card was also a region landmark, which crowded the landmark list (SR-6, fixed in PR #88: cards are no longer named regions, and only the three page sections remain). Evidence: screen reader report rows 2.3, 2.4 and 7.7.
1.3.2 Meaningful Sequence	**Supports**	Reading order and Tab order follow the visual layout on every page (keyboard checklist row 1.5). VoiceOver read Home in visual order: the heading, "3 of 10 things done today", the Next thing to do card, then Later today. On Medicines: the search, then each medicine card and its taken button.
1.3.3 Sensory Characteristics	**Supports**	Instructions refer to controls by name, not by shape, size, position or sound (review of on-screen text, rechecked in the keyboard checklist, #61).
1.4.1 Use of Color	**Supports**	State is given in words as well as color: VoiceOver reads each medicine with its state ("taken" or "not yet taken"), schedule items say done or not done, and badges use words ("Coming up", "Today").
1.4.2 Audio Control	**Not Applicable**	The app plays no audio automatically.
2.1.1 Keyboard	**Supports**	All pages, the native menu bar (Alt on Windows; Ctrl+F2 or VO+M on macOS), dialogs and forms work from the keyboard, and the main commands have shortcuts (Ctrl on Windows, Cmd on macOS: 1 to 6, N, Shift+N, P, F, Enter, Shift+S, Shift+U, Shift+C, Shift+H, =, -, 0 and /, plus F1 for help). Keyboard-only checklist on the final build (Antonio Wilson, PR #75): 38 pass, 6 fail; the keyboard fixes in PR #76 (#70, #73, #74) were re-tested and pass. Edit > Open Medication focuses the selected medicine and announces it (#71, fixed in PR #96 and re-checked with VoiceOver). Escape also closes the Add forms from the error summary after a failed save (SR-4, fixed in PR #79; ManageForms.test.tsx). Evidence: desktop-keyboard-checklist.md.
2.1.2 No Keyboard Trap	**Supports**	The Skip and Delete dialogs keep focus inside while open (VoiceOver: Tab cycles "Cancel", "Skip dose", "Close dialog") and close with Escape; the Keyboard Shortcuts window closes with Escape. No trap was found in the keyboard checklist (row 6.1) or in the NVDA and VoiceOver passes.
2.1.4 Character Key Shortcuts	**Supports**	Every shortcut uses Ctrl or Cmd, Alt or a function key; there are no single-character shortcuts (keyboard checklist row 6.2).
2.2.1 Timing Adjustable	**Supports**	No time limits: there is no session timeout, and Undo Last Dose Change (Ctrl/Cmd+Shift+U) has no time limit.

Criteria	Conformance Level	Remarks and Explanations
2.2.2 Pause, Stop, Hide	**Supports**	No moving, blinking or scrolling content. The header clock updates once a minute as part of the app's orientation purpose and is not re-announced to screen readers. Status messages stay until the next action or page change.
2.3.1 Three Flashes or Below Threshold	**Supports**	No flashing content.
2.4.1 Bypass Blocks	**Supports**	"Skip to main content" is the first Tab stop on the landing page and every signed-in screen, and moves focus to the main landmark (VoiceOver: "link, Skip to main content", then "Main content, main"; a11y.test.tsx). It stays readable in High Contrast mode (white on #0f5070, 8.74:1). Banner, navigation and main landmarks are exposed.
2.4.2 Page Titled	**Supports**	The window title names the current page as "<page> - CareConnect" (VoiceOver read "Medicines - CareConnect, window" and "My Appointments - CareConnect, window"). Sign in, Sign up and Choose your role have their own titles, the title resets on sign-out (#55, PR #58; a11y.test.tsx), and the Keyboard Shortcuts and About windows have their own titles. The caregiver Dashboard window title now reads "Dashboard - CareConnect" to match its heading (SR-7, fixed in PR #87).
2.4.3 Focus Order	**Partially Supports**	Tab and Shift+Tab follow a logical order on every page. When the Skip dialog closes, focus returns to the control that had it (confirmed with VoiceOver) or to the main content. New Medication and New Appointment put focus in the first field, and focus returns to the app when the Keyboard Shortcuts window closes (#70, #73, PR #76, re-tested). After a medicine is deleted, focus moves to the next medicine, or to the Medicines heading when none are left (#72, fixed in PR #96 and re-checked with VoiceOver). Exceptions: after the Skip dialog closes, the returned focus is not always visible (#46, open), and after confirming a delete on the caregiver Manage Medications and Manage Appointments screens, focus falls to the page body (found 11/3/2026 in the web build that Electron loads; no issue filed). The Skip dose and delete dialogs now open on Cancel, the safe choice (SR-8), and the Keyboard Shortcuts window puts focus on its heading when it opens (SR-9); both fixed in PR #87. Evidence: ConfirmDialog.test.tsx, ManageForms.test.tsx, the keyboard checklist and the screen reader report.
2.4.4 Link Purpose (In Context)	**Supports**	Links have descriptive names, for example "Skip to main content", "See full day schedule", "View contacts" and the navigation links (VoiceOver Links rotor). The Call my caregiver link names the caregiver and the number.
2.5.1 Pointer Gestures	**Supports**	No multipoint or path-based gestures.
2.5.2 Pointer Cancellation	**Supports**	Controls are native buttons and links, activated on the up-event.

Criteria	Conformance Level	Remarks and Explanations
2.5.3 Label in Name	**Supports**	Controls use their visible text as the accessible name. The three Add buttons whose names did not contain their visible text were fixed in PR #58 (#56). The header Switch role button, named "Switch to caregiver role" on 0ff70f6 (SR-10), is now "Switch role to caregiver" or "Switch role to care recipient", so its name starts with the visible "Switch role" (PR #79; VoiceOver read "Switch role to care recipient, button"; a11y.test.tsx).
2.5.4 Motion Actuation	**Not Applicable**	No functions are triggered by device motion.
3.1.1 Language of Page	**Supports**	<html lang="en"> is set in apps/web/index.html, and the Keyboard Shortcuts window sets lang="en" as well.
3.2.1 On Focus	**Supports**	Moving focus never changes context (keyboard checklist).
3.2.2 On Input	**Supports**	Changing a field or setting does not change context. Edit > Find keeps a half-written caregiver note and says why (#38).
3.3.1 Error Identification	**Supports**	Submitting an empty Add Medication form moves focus to the error summary, which VoiceOver reads as an alert: "Please fix the following before saving:, and 1 more item, alert". It does not go on to read the errors themselves; VO+Right moves into the list, which names each missing field. Each field also shows its own error, linked with aria-describedby and read when the field gets focus (screen reader report, section 12).
3.3.2 Labels or Instructions	**Supports**	Every form field has a visible label, and required fields are marked in the label and programmatically (the required attribute). VoiceOver reads the visible "(required)" once and then the field's required state ("Medication name (required) e.g. Amlodipine, required, edit text"); PR #79 removed a hidden duplicate that made it three times (SR-2). Minor wording mismatch: the field says "Dose" and the error says "Dosage is required".
4.1.1 Parsing	**Supports**	The interface is rendered by React as well-formed DOM. (This criterion is obsolete in WCAG 2.2 and always satisfied for HTML.)
4.1.2 Name, Role, Value	**Supports**	Controls expose name, role and state: NVDA read "File, button, collapsed"; VoiceOver read "Sign out of CareConnect, button", "current page, link, Medicines" and toggle states as "selected". aria-labels on plain spans and divs, which screen readers ignore, were replaced with visible or visually hidden text (PR #69). axe ARIA and name rules: 0 violations on 18 screens (#63). Fixed in PR #88: a new medication time now starts with a value, so Chromium no longer reads an empty time field as steppers with meaningless values (SR-3), and the taken toggles keep one fixed name and report their state only through aria-pressed ("Mark Amlodipine taken, toggle button", SR-5).

Table 2: Success Criteria, Level AA

Criteria	Conformance Level	Remarks and Explanations
1.2.4 Captions (Live)	**Not Applicable**	The app contains no live audio or video.

Criteria	Conformance Level	Remarks and Explanations
1.2.5 Audio Description (Prerecorded)	**Not Applicable**	The app contains no prerecorded video.
1.3.4 Orientation	**Supports**	No orientation lock; the window can be resized and the layout adapts (wide and narrow layouts).
1.3.5 Identify Input Purpose	**Supports**	Sign-in and sign-up fields set autocomplete (email, current-password, name, new-password).
1.4.3 Contrast (Minimum)	**Supports**	All 14 text colour tokens are at least 4.5:1 on white and neutral-50 (npm run check:contrast, run in CI). Secondary text moved from neutral-400 (2.85:1) to neutral-500 (5.53:1) in 32 places (#57, PR #58). axe color-contrast in the Electron window: 0 violations on 18 screens in normal and High Contrast mode (dev 0ff70f6, #63).
1.4.4 Resize Text	**Supports**	Zoom In (Ctrl/Cmd+=) to about 200% keeps every page usable with no cut-off content; Actual Size (Ctrl/Cmd+0) resets.
1.4.5 Images of Text	**Supports**	No images of text.
1.4.10 Reflow	**Supports**	At a 320 CSS px wide viewport (400% zoom of a 1280 px window), none of the 12 pages tested scrolls horizontally; content reflows to one column (Electron window, 10/9/2026).
1.4.11 Non-text Contrast	**Supports**	Borders and the focus ring (#146488) are at least 3:1 on white and neutral-50 (npm run check:contrast). The focus ring on the red "Yes, delete" button meets 3:1 (#53, PR #58). High Contrast mode (View > High Contrast) restyles text, borders, fills and the focus ring, with every High Contrast colour at least 7:1 (#44, PR #58; high-contrast.test.tsx). The Keyboard Shortcuts help window does not switch to High Contrast; its normal colours pass axe.
1.4.12 Text Spacing	**Supports**	With the WCAG text-spacing values applied, no text is cut off or overlaps on 12 pages (Electron window, 10/9/2026). Memories cards show an intentional three-line preview, and Read more reveals the full text.
1.4.13 Content on Hover or Focus	**Supports**	The only hover content is native title tooltips, which the platform controls; there are no custom hover popups.
2.4.5 Multiple Ways	**Supports**	Pages are reachable from the navigation, the View menu and Ctrl/Cmd+1 to 6.
2.4.6 Headings and Labels	**Supports**	Headings and labels describe their topic or purpose (VoiceOver headings rotor; NVDA in A8).
2.4.7 Focus Visible	**Supports**	A visible outline appears on every focusable control (keyboard checklist rows 1.6 and 6.3, including High Contrast mode, where the ring is a 4 px black outline). The ring on the red "Yes, delete" button meets 3:1 (#53, PR #58). The one gap, focus after the Skip dialog closes, is under 2.4.3 (#46). A focused control can also be covered by the fixed bottom navigation; see Known Limitations.
3.1.2 Language of Parts	**Not Applicable**	All content is in English.
3.2.3 Consistent Navigation	**Supports**	The navigation and the menu bar keep the same order on every page.

Criteria	Conformance Level	Remarks and Explanations
3.2.4 Consistent Identification	**Supports**	The demo patient is "Margaret" on every screen (#48, PR #58), and every View menu item matches the page it opens (#47, PR #54; Medicines, PR #60).
3.3.3 Error Suggestion	**Supports**	The error summary lists each missing field (name, dosage, time).
3.3.4 Error Prevention (Legal, Financial, Data)	**Supports**	Deleting a medication and skipping a dose ask for confirmation, and dose changes can be undone (Ctrl/Cmd+Shift+U).
4.1.3 Status Messages	**Supports**	Mark Next Dose Taken, Undo, Remind Me and Skip announce through a polite live region without moving focus: VoiceOver spoke "Evening medications marked as taken." and "Undid the last change to Evening medications.", and NVDA spoke the same (A8). The medicine search result is announced once after typing pauses (#40, #38).

Table 3 (Level AAA) is not included in this report. The Web report rates selected Level AAA criteria for the same web code.

Known Limitations and Workarounds

Open issues a user can still meet in this build, with a workaround for each:

Issue	Limitation	Workaround
#46	After the Skip dialog closes, keyboard focus is not always visibly returned (2.4.3). VoiceOver focus did return to the previous control. GitHub closed #46 automatically when PR #89 merged; the limitation remains.	Press Tab to move focus back into the page.
#52	Badge text runs into the next word in NVDA ("Coming upAppointment") on Home and Appointments (1.3.1).	Read by word (NVDA Ctrl+Right Arrow). VoiceOver reads them as separate items with VO+Right Arrow.
SR-8	Confirmation dialogs open on Cancel, but VoiceOver does not read the question until you move to it (4.1.2).	Press VO+Right to hear the question.
Empty form errors (no issue filed)	Saving an empty form moves focus to the error summary, but VoiceOver reads only its heading and "and 1 more item", not the errors themselves (3.3.1). The rating stays Supports because the errors are identified in text and reachable.	Press VO+Right to move into the list, or Tab to a field, which reads its own error.
Delete focus (no issue filed)	After confirming a delete on Manage Medications or Manage Appointments, focus falls to the page body, and nothing is announced (2.4.3).	Press Tab: focus continues on the next item's Edit button.
Bottom navigation (no issue filed)	A focused control at the bottom of the window can be completely covered by the fixed bottom navigation on the care recipient screens (WCAG 2.2, 2.4.11, which this WCAG 2.1 report does not rate).	Scroll with the arrow keys or Page Down to bring the control above the navigation.
Keyboard Shortcuts window (no issue filed)	The Keyboard Shortcuts help window does not switch to High Contrast.	Use the menu bar, which lists every command with its shortcut.

Fixed and merged: #44, #48, #49, #53, #55, #56 and #57 (PR #58); #45, #47 and #51 (PR #54); #50 (PR #59, with tests in PR #68); the View menu Medicines label (PR #60); ignored aria-labels on plain elements (PR #69); #70, #73 and #74 (PR #76); SR-1, SR-2, SR-4 and SR-10 (PR #79); SR-7 and SR-9, and the Cancel focus part of SR-8 (PR #87); SR-3, SR-5 and SR-6 (PR #88); #71 and #72 (PR #96); and the spoken medicines count (PR #102).

Evidence

- Screen reader report: docs/accessibility/desktop-screen-reader-report.md, with VoiceOver caption screenshots in docs/accessibility/screen-reader/ and the 11/3/2026 log in screen-reader/retest-2026-11-03.json. Videos: Team5_A9_Desktop_VoiceOver_Screen_Reader_Test.mp4 (3:51) and SWEN661_Team5_A12_Keyboard_VoiceOver.mp4 (3:44, the same web code in Chrome). NVDA speech log from the A8 pass (10/6/2026).
- Keyboard-only checklist: docs/accessibility/desktop-keyboard-checklist.md (PR #75): 38 pass, 6 fail on dev 8a7a1a5, tester Antonio Wilson; #70, #73 and #74 re-tested after PR #76.
- axe-core: docs/accessibility/axe/ (PR #78): axe-core 4.14.0, 0 violations in 36 scans (18 screens, normal and High Contrast mode) on dev 0ff70f6, with a results panel on every screenshot, the JSON results and a README on the needs-review items.
- Reflow and text-spacing results and screenshots for 12 pages (10/9/2026), and High Contrast spot checks: skip link 8.74:1, Landing hero 6.55 to 8.74:1, Appointments "Today" 12.05:1.
- Coverage reports: apps/web/coverage/ and apps/desktop/coverage/ in the repository.

Open Items

- When #46 and #52 are fixed and re-tested, 2.4.3 and 1.3.1 move to Supports.
- File issues for the delete focus problem, the bottom navigation covering focus, the empty form error summary, and the Keyboard Shortcuts window not switching to High Contrast.

Revision History

Version	Date	Author	Changes
1.1	10/13/2026	Quinton Coleman	Every WCAG 2.1 A and AA criterion rated; keyboard-only checklist results from PR #75.
1.2	10/13/2026	Shayne McPherson	VoiceOver pass (#62) added to every relevant row; #70, #73 and #74 marked fixed (PR #76); #46 taken out of the fixed list; tests re-run on dev 0ff70f6; 2.5.3 set to Partially Supports (SR-10, fixed in PR #79); evidence cited by repository path or issue.
1.3	10/13/2026	Shayne McPherson	Rated with PR #79 merged: 2.5.3 now Supports, and SR-1, SR-2, SR-4 and SR-10 marked fixed; axe-core re-run on 18 screens in both modes (0 violations, docs/accessibility/axe/); tests 119/119 web, 63/63 desktop.

Version	Date	Author	Changes
1.4	11/1/2026	Quinton Coleman	Added PRs #87 and #88, which fix SR-3 and SR-5 to SR-9 (automated tests; VoiceOver re-check still to do). Noted that #46 was closed automatically by PR #89 but is still a known limitation. Ratings unchanged.
1.5	11/3/2026	Shayne McPherson	Added PR #96, which fixes #71 and #72, and a VoiceOver re-check of SR-3, SR-5 to SR-9, #71 and #72. SR-8 is partly fixed. 2.1.1 moved to Supports: 40 Supports, 2 Partially Supports, 8 Not Applicable.
1.6 (Final)	11/3/2026	Shayne McPherson	Final build and PRs through #102 recorded. 3.3.1 corrected to say how VoiceOver reads the error summary. Three limitations added from the 11/3 evaluation of the web build: the error summary, focus after a delete on the Manage screens, and focus covered by the bottom navigation. Open items updated. Ratings unchanged: 40 Supports, 2 Partially Supports, 8 Not Applicable.

Legal Disclaimer

This report was prepared by SWEN 661 Team 5 (UMGC, Fall 2026) for coursework. It describes CareConnect Desktop as tested on the dates above and is not a legal statement of conformance.